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systemd system is a startup system but also a service manager for Linux, while the system is running.

Exercise with systemd commands:

- Apache web server status task:

Check the status of the httpd (Apache Web-Server) server on your system.

- if running, stop it and check again
- start the service and check status
- stop it again when done

```
$ sudo systemctl status httpd
$ sudo systemctl start httpd
$ sudo systemctl status httpd
$ sudo systemctl stop httpd
```

## LINUX Filesystems

Libraries organize books and other media by subject, audience, media type, frequency, retrieval.

Same goes for Linux file system = a method of storing and organizing arbitrary collections of data.

Types of filesystems supported by Linux

- disk file systems
  - ext3, ext4, XFS, Btrfs etc
- flash storage file systems
  - UFS, JFS, etc



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init program to complete the startup process.

- Distro needs to match your specific system needs.



## When Choosing a Distribution:

- what is the main function of the system?  
(server / desktop)
- what type of packages are important to the organization?  
(web server, web word processing etc)
- Storage needed
- how often are packages updated?
- support cycle for each release
- what hardware are you running on?  
(x86, RISC-V, ARM, PPC, etc)
- long-term stability or a more volatile cutting-edge system running the latest software versions?

## Key Concepts

- **partition** = logical part of the disk
- **filesystem** = method of storing / finding files on a hard disk.
- By dividing the **hard disk** into **partitions**, data can be grouped and separated as needed. When a failure or mistake occurs, only the data in the affected partition will be damaged.
- **Boot process**: BIOS (triggers the **boot loader** to start up the **linux kernel**), then **initramfs** filesystem is invoked (this triggers the

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- mnt / media  $\Rightarrow$  used for mounting or attaching other file systems.
- etc  $\Rightarrow$  directory where system wide configuration files are stored.
- var  $\Rightarrow$  where changeable or variable system info is kept. System logs and logs for other software.
- dev / proc / sys directories are created by the kernel to represent hardware available on the system including all the systems hardware, processes that run programs, settings in the kernel etc.

- Linux uses the '/' character to separate paths.
- All Linux filesystem names are case-sensitive, so /boot, /Boot and /BOOT represent 3 different directories / folders.

How to view filesystem hierarchy from GUI

- applications  $\rightarrow$  files
- or
- activities and it will show at the bottom

## Linux Distributions

### Server

- RHEL / CentOS
- Ubuntu Server
- SLES / openSUSE

### Desktop

- Ubuntu
- Fedora
- Debian

### Embedded

- Yocto
- Open Embedded
- Android



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- database file systems
- special purpose file systems  
procs, sysfs, debugfs, fuse, etc

## Standard file system layout

Files and directories make up the file system.  
(aka folders on other OSs)

FHS = filesystem hierarchy standard

- helps to ensure consistent locations across distributions

- The TOP LEVEL of the file system is called the root represented by / (slash)

- Everything starts with the file system root in linux.  
There's only one file system root. (ms C, Δ etc)

File system is like "My Computer" on Windows.

- From the root we move deeper into the file system.  
1st level - variety of directories with specific purpose

Some of the important directories defined by the

FHS:

- home => where each user's personal files are stored.
- bin / sbin / usr => where programmes of different types are kept.

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## startup alternative

- systemd = modern startup (init) system for Linux
- it replaces older init systems like SysVinit and Upstart that started services step by step.

## \* systemd

- boots up the system faster by starting many services in parallel.
- uses configuration files instead of long, messy shell scripts.
- manages not only startup, but also services while the system is running

## How to use:

- with the

systemctl command

You can start, stop, restart services, enable or disable them at boot, and check status.

## Big chance

Old /sbin/init now just points to systemd.

systemd has taken over the role of the init process.

↓  
The process that starts when Linux boots.

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